Use-Case Flow Specification

Use Case: Ingest and Deliver Webhook

Primary Actor: Integration Developer
Supporting Actor: System Operator

1. Preconditions

• The webhook gateway is available to receive inbound webhook events.

• A target internal service endpoint is configured for the incoming webhook.

• The system is able to record request payloads and delivery activity.

2. Postconditions

Success: The webhook payload has been forwarded to the target internal service and the request/delivery activity has been logged.

Failure after retries: The webhook delivery has been retried using the specified exponential backoff sequence and, if still unsuccessful, the webhook has been persisted in the dead-letter queue.

3. Main Success Scenario

1. The Integration Developer's third-party integration sends an inbound webhook to the gateway.
2. The gateway accepts the inbound webhook.
3. The gateway logs the received request payload.
4. The gateway identifies the configured internal target endpoint.
5. The gateway forwards the webhook payload to the target endpoint.
6. The target endpoint successfully receives the payload.
7. The gateway records the successful delivery.
8. The use case completes successfully.

4. Alternate Flow — Delivery Failure and Retry

1. At Step 5, the target endpoint fails to accept the webhook.
2. The gateway records the failed delivery.
3. The gateway retries delivery after 1 second.
4. If delivery still fails, the gateway retries after 2 seconds.
5. If delivery still fails, the gateway retries after 4 seconds.
6. If delivery still fails, the gateway retries after 8 seconds.
7. If delivery remains unsuccessful after the retry process, the gateway persists the webhook in the dead-letter queue.
8. The System Operator can subsequently manage the dead-lettered webhook.

5. Actors

Integration Developer: Initiates/uses the webhook integration and is the primary stakeholder for successful event delivery.

System Operator: Operates the gateway and manages failed/dead-lettered webhook deliveries.